Multiple linear regression

We study the relationship between a response variable $Y \in \mathbb{R}$ and a covariate (predictors, explanatory variables, feature) $\underline{X} \in \mathbb{R}^p$.

$$Y = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

We assume that the conditional distribution of Y given $\underline{X} = \underline{x}$ has the form

$$Y \mid \underline{X} = \underline{x} \sim N(\underline{x}\underline{\beta}, \sigma^2)$$

$\underline{x} = (x_1, \ldots, x_p)$

$\underline{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$

Often we take $x_1 = 1$ so here β_1 is the "intercept" of the model.

Assume our data is of the form

$$(Y_1, \underline{X}_1), (Y_2, \underline{X}_2), \ldots, (Y_n, \underline{X}_n) \overset{IID}{\sim} F_{Y,\underline{X}}$$

Suppose $(Y, \underline{X})$ has a joint pdf $f_{Y,\underline{X}}$. Then the likelihood function is

$$\begin{aligned} \mathcal{L}(\underline{\beta}, \sigma^2) &= \prod_{j=1}^{n} f_{Y,\underline{X}}(y_j, \underline{x}_j ; \underline{\beta}, \sigma^2) \\ &= \prod_{j=1}^{n} f_{Y|\underline{X}}(y_j \mid \underline{x}_j ; \underline{\beta}, \sigma^2) \, f_{\underline{X}}(\underline{x}_j) \\ &= \underbrace{\left\{ \prod_{j=1}^{n} f_{Y|\underline{X}}(y_j \mid \underline{x}_j ; \underline{\beta}, \sigma^2) \right\}}_{\mathcal{L}_1 \text{ – conditional likelihood}} \underbrace{\left\{ \prod_{j=1}^{n} f_{\underline{X}}(\underline{x}_j) \right\}} \end{aligned}$$

The conditional likelihood is

$$\mathcal{L}_1(\underline{\beta}, \sigma^2) = \prod_{j=1}^{n} (2\pi\sigma^2)^{-1/2} \cdot \exp\left(\frac{-1}{2\sigma^2} (y_j - \underline{x}_j\underline{\beta})^2 \right)$$

The conditional log likelihood is

$$\ell(\underline{\beta}, \sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^{n} (y_j - \underline{x}_j\underline{\beta})^2 - \frac{n}{2}\log(2\pi)$$

It is useful to write this in matrix form

$$\underline{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \qquad \underline{\underline{X}} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ & & & \\ & & & \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

← model matrix or Design matrix.

Then

$$\sum_{j=1}^{n} (y_j - x_j \beta)^2 = (\underline{y} - \underline{\underline{X}}\underline{\beta})^T (\underline{y} - \underline{\underline{X}}\underline{\beta}).$$

We can differentiate wrt β and set this equal to 0. The MLE of β satisfies the equation

$$\underline{\underline{X}}^T \underline{\underline{X}}\, \hat{\underline{\beta}} = \underline{\underline{X}}^T \underline{y}. \qquad \text{normal equations.}$$

Provided $\underline{\underline{X}}^T\underline{\underline{X}}$ is invertible, the MLE is unique and given by

$$\hat{\underline{\beta}} = (\underline{\underline{X}}^T\underline{\underline{X}})^{-1} \underline{\underline{X}}^T \underline{y}. \qquad (*)$$

Take $\underline{u} \in \mathbb{R}^p$ and $h \in \mathbb{R}$.

$$(\underline{y} - \underline{\underline{X}}(\hat{\underline{\beta}} + h\underline{u}))^T (\underline{y} - \underline{\underline{X}}(\hat{\underline{\beta}} + h\underline{u}))$$

$$= (\underline{y} - \underline{\underline{X}}\hat{\underline{\beta}})^T (\underline{y} - \underline{\underline{X}}\hat{\underline{\beta}}) - 2h\, \underline{u}^T \underline{\underline{X}}^T (\underline{y} - \underline{\underline{X}}\hat{\underline{\beta}}) + h^2\, \underline{u}^T \underline{\underline{X}}^T \underline{\underline{X}}\, \underline{u}$$

The value of h which minimises this is

$$\frac{\underline{u}^T \underline{\underline{X}}^T (\underline{y} - \underline{\underline{X}}\hat{\underline{\beta}})}{\underline{u}^T \underline{\underline{X}}^T \underline{\underline{X}}\, \underline{u}} = 0$$

show $\underline{u}^T\underline{\underline{X}}^T\underline{\underline{X}}\,\underline{u} > 0$ for all $\underline{u} \in \mathbb{R}^p$.

since $\underline{\underline{X}}^T \underline{y} = \underline{\underline{X}}^T \underline{\underline{X}}\, \hat{\underline{\beta}}$ from the normal equations. So $\hat{\underline{\beta}}$ does minimise the sum of squares.

$\underline{\underline{X}}^T\underline{\underline{X}}$ is invertible if $\underline{\underline{X}}$ is of full rank, ie. the p columns of $\underline{\underline{X}}$ are linearly independent.

Having minimised the sum of squares, the MLE of σ^2 can be found by minimising

$$\ell(\hat{\beta}, \sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(y - \underline{\underline{X}}\hat{\beta})^T(y - \underline{\underline{X}}\hat{\beta})$$

Following the same calculation as for the $N(\mu, \sigma^2)$ MLE, we find

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}(y - \underline{\underline{X}}\hat{\beta})^T(y - \underline{\underline{X}}\hat{\beta}) = \frac{1}{n}\sum_{j=1}^{n}(y_j - \underline{x}_j\hat{\beta})^2.$$

We usually use the unbiased estimator

$$\hat{\sigma}^2_{unbiased} = \frac{1}{n-p}(y - \underline{\underline{X}}\hat{\beta})^T(y - \underline{\underline{X}}\hat{\beta}).$$

STAT2003 Claim 45: Suppose V has mean μ_V and covariance matrix Σ_V. Let $W = AV$ where A is a constant matrix. Then

$$\mu_W = A\mu_V \quad \text{and} \quad \Sigma_W = A\Sigma_V A^T.$$

Let's write $\underline{Y} = (Y_1, \ldots, Y_n)^T$. and (conditional on $\underline{\underline{X}}$)

$$\underline{Y} \sim N(\underline{\underline{X}}\beta, \sigma^2 I) \qquad \left(\begin{matrix}\text{multivariate}\\ \text{normal}\\ \text{distribution}\end{matrix}\right)$$

(1) $\hat{\beta}$ is an unbiased estimator of β.

Proof: $E(\hat{\beta}) = E\left((\underline{\underline{X}}^T\underline{\underline{X}})^{-1}\underline{\underline{X}}^T\underline{Y}\right) = (\underline{\underline{X}}^T\underline{\underline{X}})^{-1}\underline{\underline{X}}^T E(Y)$

$$= (\underline{\underline{X}}^T\underline{\underline{X}})^{-1}\underline{\underline{X}}^T\underline{\underline{X}}\beta = \beta \qquad \square$$

(2) The covariance matrix of $\hat{\beta}$ is $\sigma^2(\underline{X}^T\underline{X})^{-1}$.

Proof: $\text{cov}(\hat{\beta}) = \text{cov}\left((\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{Y}\right) = (\underline{X}^T\underline{X})^{-1}\underline{X}^T \,\text{cov}(\underline{Y})\left((\underline{X}^T\underline{X})^{-1}\underline{X}^T\right)^T$

$= (\underline{X}^T\underline{X})^{-1}\underline{X}^T(\sigma^2 I)\underline{X}(\underline{X}^T\underline{X})^{-1}$

note $\underline{X}^T\underline{X}$ is symmetric so is $(\underline{X}^T\underline{X})^{-1}$.

$= \sigma^2(\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{X}(\underline{X}^T\underline{X})^{-1}$

$= \sigma^2(\underline{X}^T\underline{X})^{-1}$ □

(3) $\hat{\beta} \sim N(\beta, \sigma^2(\underline{X}^T\underline{X})^{-1})$

Proof: We have determined expectation and covariance matrix of $\hat{\beta}$ in (1) and (2). As $\underline{Y} \sim N(\underline{X}\beta, \sigma^2 I)$ and $\hat{\beta}$ is a linear transformation of $\underline{Y}$, $\hat{\beta}$ has a (multivariate) normal distribution.

QR decomposition and distribution of $\hat{\sigma}^2_{\text{unbiased}}$.

Any matrix $\underline{X}$ ← $n \times p$ matrix can be decomposed as

$$\underline{X} = Q\begin{bmatrix} R \\ 0 \end{bmatrix}$$

R: $p \times p$ upper triangular.

0: $(n-p) \times p$ zero matrix

Q: $n \times n$ orthogonal matrix

$Q Q^T = Q^T Q = I$.

$$Q^T\underline{X} = Q^T Q\begin{bmatrix} R \\ 0 \end{bmatrix} = \begin{bmatrix} R \\ 0 \end{bmatrix}$$

Since Q^T is orthogonal, it doesn't change the length of vectors.

Applying Q^T to $y - \underline{X}\beta$

$$\|y - \underline{X}\beta\|^2 = \|Q^T y - Q^T\underline{X}\beta\|^2 = \left\| Q^T y - \begin{bmatrix} R \\ 0 \end{bmatrix}\beta \right\|^2$$

write $Q^T y = \begin{bmatrix} f \\ r \end{bmatrix}$ ← dimension p

← dimension $n-p$.

Then

$$\|y - X\beta\|^2 = \left\| \begin{bmatrix} f \\ r \end{bmatrix} - \begin{bmatrix} R \\ 0 \end{bmatrix} \beta \right\|^2 = \|f - R\beta\|^2 + \|r\|^2$$

So the MLE $\hat{\beta}$ is $R^{-1}f$ and then $\|r\|^2$ is the sum of squares (residuals) - it does not depend on β.

What is the distribution of $\begin{bmatrix} f \\ r \end{bmatrix}$?

$$E\begin{bmatrix} f \\ r \end{bmatrix} = E(Q^T Y) = Q^T EY = Q^T X\beta = \begin{bmatrix} R\beta \\ 0 \end{bmatrix}$$

$$\text{Cov}\begin{bmatrix} f \\ r \end{bmatrix} = \text{cov}(Q^T Y) = Q\,\text{cov}(Y)\,Q^T = Q(\sigma^2 I)Q^T$$

$$= \sigma^2 QQ^T = \sigma^2 I$$

So $$\begin{bmatrix} f \\ r \end{bmatrix} \sim N\left(\begin{bmatrix} R\beta \\ 0 \end{bmatrix}, \sigma^2 I\right).$$

In particular, $f \sim N(R\beta, \sigma^2 I_p)$, $r \sim N(0, \sigma^2 I_{n-p})$ and f and r are independent. (since they are normally distributed and uncorrelated).

So the $r_j \sim N(0, \sigma^2)$ and they are independent of each other. Hence

$$\frac{1}{\sigma^2}\|r\|^2 = \frac{1}{\sigma^2}(r_1^2 + r_2^2 + \cdots + r_{n-p}^2) \sim \chi^2_{n-p}.$$

Note that $\hat{\sigma}^2_{\text{unbiased}} = \frac{1}{n-p}\|r\|^2$. As χ^2_k has mean k. So $E\,\hat{\sigma}^2_{\text{unbiased}} = E\left(\frac{1}{n-p}\|r\|^2\right) = \frac{1}{n-p} \cdot \sigma^2 \cdot n-p = \sigma^2$.

ALSO $\hat{\sigma}^2_{\text{unbiased}} \sim$

$$(n-p)\,\frac{\hat{\sigma}^2_{\text{unbiased}}}{\sigma^2} \sim \chi^2_{n-p}.$$

Since $\hat{\beta}_j \sim N(\beta_j, \sigma^2 (\underline{X}^T\underline{X})^{-1}_{jj})$

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\text{unbiased}}\sqrt{(\underline{X}^T\underline{X})^{-1}_{jj}}} = \frac{(\hat{\beta}_j - \beta_j)/\sigma\sqrt{(\underline{X}^T\underline{X})^{-1}_{jj}}}{\sqrt{\frac{1}{\sigma^2}\|\underline{r}\|^2/(n-p)}}$$

$\sim N(0,1)$

$\sim \sqrt{\chi^2_{n-p}/n-p}$

This leads to $\dfrac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\text{unbiased}}\sqrt{(\underline{X}^T\underline{X})^{-1}_{jj}}} \sim t_{n-p}$ distribution.